

DR.BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination Summer 2024

Course: B.Tech

Semester: V

Branch: Mechanical / Automobile / Mechanical Engineering Sandwich

Subject Code & Name: BTMC506 Applied Thermodynamics

Max Marks: 60

Date: 12/07/2024

Duration: 3.00 Hr.

Instructions to the students:

1. All the questions are compulsory.
2. The level of questions/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculator is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of steam table is allowed.

(CO) Marks

Q. 1 Solve any Two of the following.

- A) List and explain the types of fuels. (CO1) 6
- B) Develop the equation for the measurement of calorific value of solid fuel by the use of bomb calorimeter with neat sketch. (CO1) 6
- C) Calculate the net and gross calorific value of a coal sample having following composition. C = 82%, H = 8%, O = 5%, N = 1.4% and ash = 3.6% (CO1) 6

Q. 2 Solve any Two of the following.

- A) Write about requirement of a good boilers. (CO3) 6
- B) Define and explain mountings and accessories with suitable any one schematic diagram of both. (CO2) 6
- C). Dry saturated steam at 5 bar with negligible velocity expands in convergent nozzle to 1 bar and dryness fraction 0.94. Determine the velocity of steam leaving the nozzle (CO1) 6

Q. 3 Solve any Two of the following.

- A) Steam expands to 2.5 bar to 1 bar in a nozzle. The initial velocity of steam is 80 m/s and initial temperature is 200°C. Taking nozzle efficiency as 90%, find the exit velocity. (CO2) 6
- B) Derive an equation of stirling cycle with neat sketch. (CO2) 6

C) Explain about super saturated flow through nozzles. (CO2) 6

Q. 4 Solve any Two of the following.

A) Write advantages and classification of steam turbines. (CO4) 6

B) List the types of cooling tower and explain any two of them with neat sketch. (CO4) 6

C) In a single stage impulse turbine, the blade angles are equal and the nozzle angle (CO4) 6

is 20° . The velocity coefficient for the blade is 0.83. Find the maximum blade efficiency possible.

If the actual blade efficiency is 90% of maximum blade efficiency, find the possible ratio of blade speed to steam speed.

Q. 5 Solve any Two of the following.

A) Explain in detail of single stage reciprocating air compressor with neat sketch. (CO5) 6

B) Write construction and working of root-blower. (CO2) 6

C) Explain the benefits of the multi staging in reciprocating air compressor with (CO2) 6

PV diagram.

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